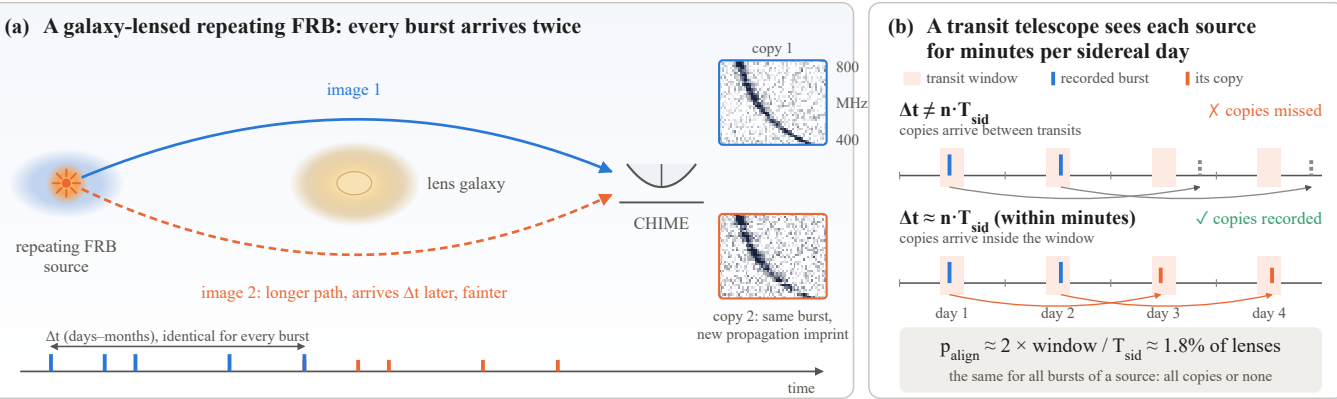

(a) A galaxy-lensed repeating FRB: every burst arrives twice
image 1
lens galaxy
repeating FRB source
CHIME
image 2: longer path, arrives Δt later, fainter
copy 1
800
MHz
400
copy 2: same burst, new propagation imprint
Δt (days–months), identical for every burst
time
(b) A transit telescope sees each source for minutes per sidereal day
transit window
recorded burst
its copy
Δt ≠ n·T_sid
copies arrive between transits
✗ copies missed
Δt ≈ n·T_sid (within minutes)
copies arrive inside the window
✓ copies recorded
day 1
day 2
day 3
day 4
p_align ≈ 2 × window / T_sid ≈ 1.8% of lenses
the same for all bursts of a source: all copies or none